1. $Na + Cl_2 \rightarrow NaCl$
2. $CaCO_3 \rightarrow CaO + CO_2$
3. $Zn + HCl \rightarrow ZnCl_2 + H_2$
4. $HCl + NaOH \rightarrow NaCl + H_2O$
5. $CH_4 + O_2 \rightarrow CO_2 + H_2O$

6. $K + Br_2 \rightarrow KBr$

7. $Fe_2O_3 \rightarrow Fe + O_2$

8. $AgNO_3 + NaCl \rightarrow NaNO_3 + AgCl$

9. $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

10. $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$

11. $H_2 + O_2 \rightarrow H_2O$

12. $Al + Cl_2 \rightarrow AlCl_3$

13. $KClO_3 \rightarrow KCl + O_2$

14. $Na_2CO_3 + HCl \rightarrow NaCl + H_2CO_3$ ($\rightarrow CO_2 + H_2O$)

15. $CH_4 + H_2O \rightarrow CO_2 + H_2O$

16. $Fe + CuCl_2 \rightarrow FeCl_2 + Cu$

17. $Ca + O_2 \rightarrow CaO$

18. $NH_4NO_3 \rightarrow N_2O + H_2O$

19. $Pb(NO_3)_2 + KI \rightarrow PbI_2 + KNO_3$

20. $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$

21. A synthesis reaction forms a more complex product from simpler reactants. TRUE
22. Decomposition reactions require heat, light, or electricity to occur. TRUE
23. Combustion reactions always involve oxygen as a reactant. TRUE
24. In a single replacement reaction, one element replaces another in a compound. TRUE
25. Double replacement reactions produce one element and one compound. FALSE

26. The reaction $NaOH + HCl \rightarrow NaCl + H_2O$ is a double replacement reaction. TRUE

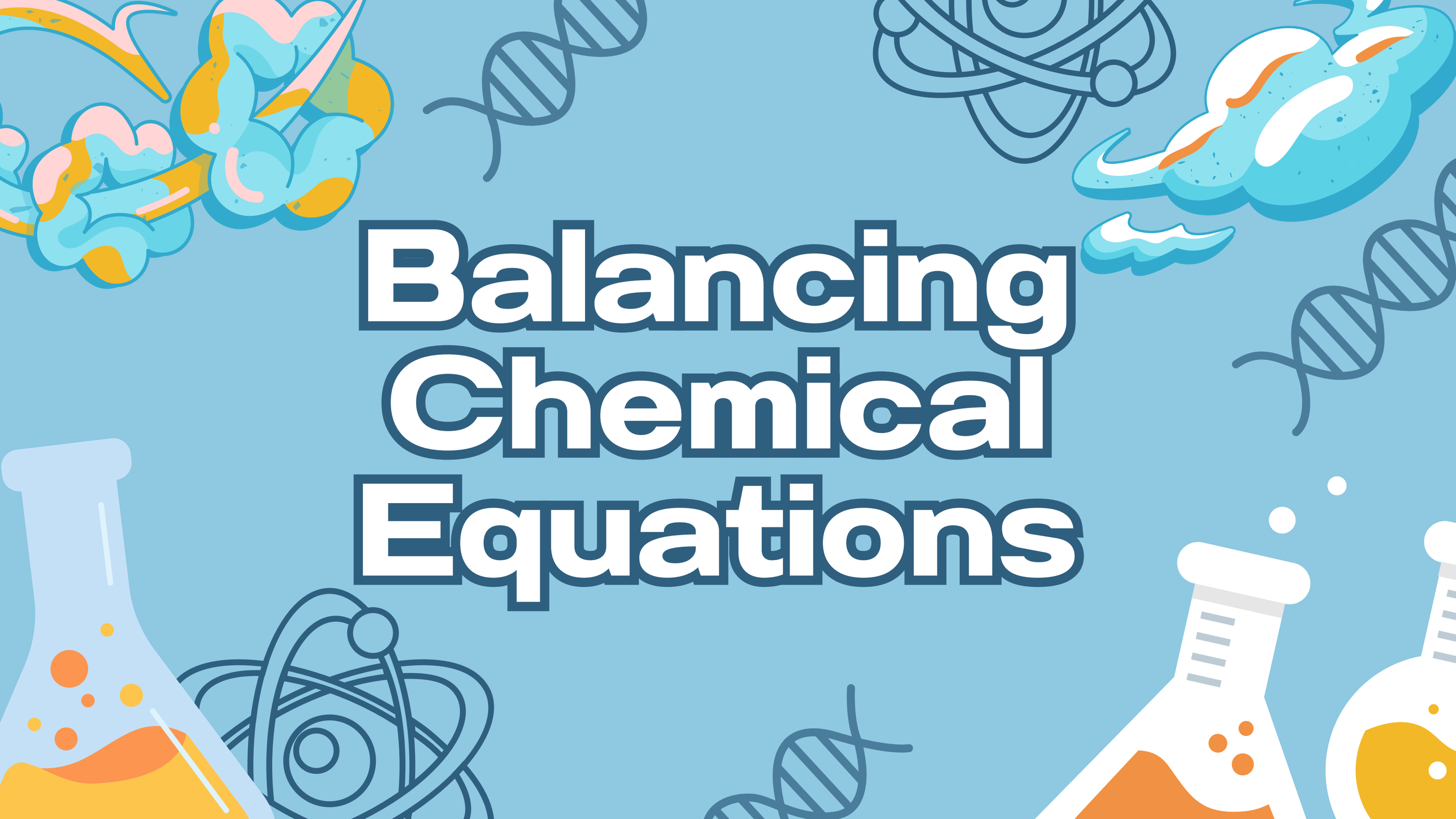
27. All decomposition reactions result in the formation of gases. FALSE

28. Combustion reactions are always exothermic (release heat). TRUE

29. A double replacement reaction usually occurs between two ionic compounds in solution. TRUE

30. In a synthesis reaction, two or more substances combine to form one new substance. TRUE

Balancing Chemical Equations



Learning Objectives:

- Identify the parts of a chemical equation (reactants, products, coefficients, and subscripts)
- Explain the Law of Conservation of Mass and how it relates to balancing chemical equations
- Demonstrate the ability to balance chemical equations involving synthesis, decomposition, single replacement, and double replacement reactions

Antoine-Laurent Lavoisier

(26 August 1743 - 8 May 1794)



Father of Modern Chemistry

- French nobleman and Chemist
- He named oxygen (1778), recognizing it as an element, and also recognized hydrogen as an element (1783)

The Law of Conservation of Mass

The Law of Conservation of Mass says:

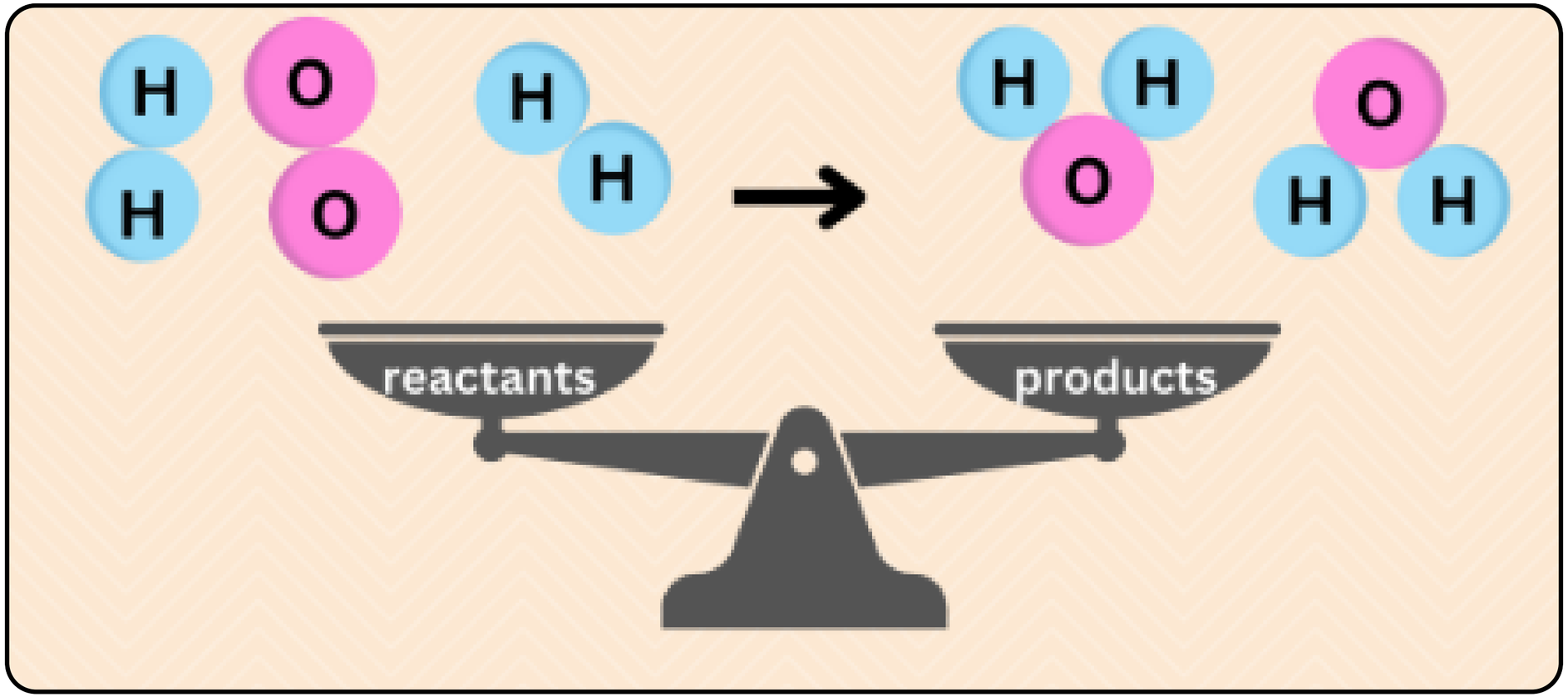
"Mass is neither created nor destroyed in a chemical reaction."

The Law of Conservation of Mass

The Law of Conservation of Mass says:

This means the total mass of the reactants is equal to the total mass of the products.

The Law of Conservation of Mass



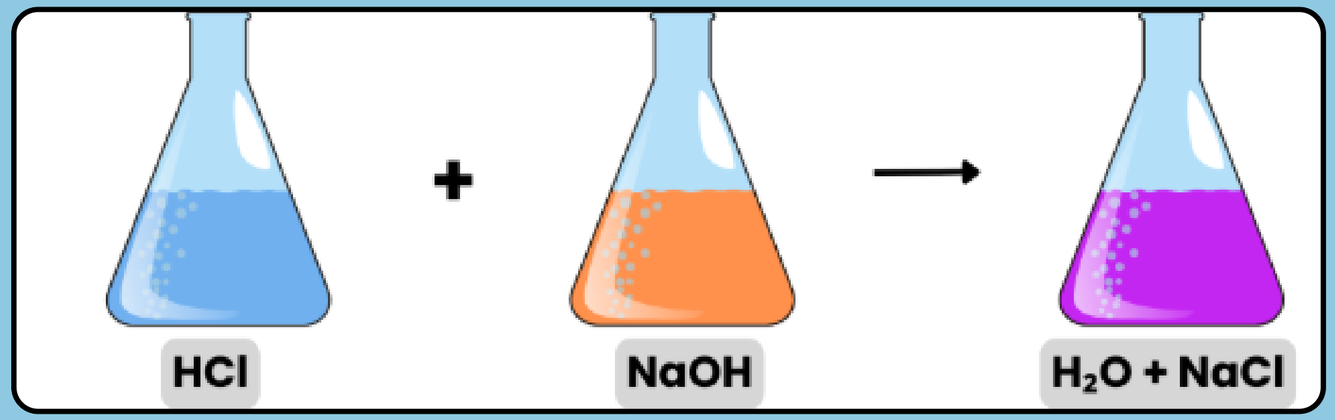
The Law of Conservation of Mass

The mass of the system is the same before and after the reaction.

The number and type of atoms does not change.

The Law of Conservation of Mass

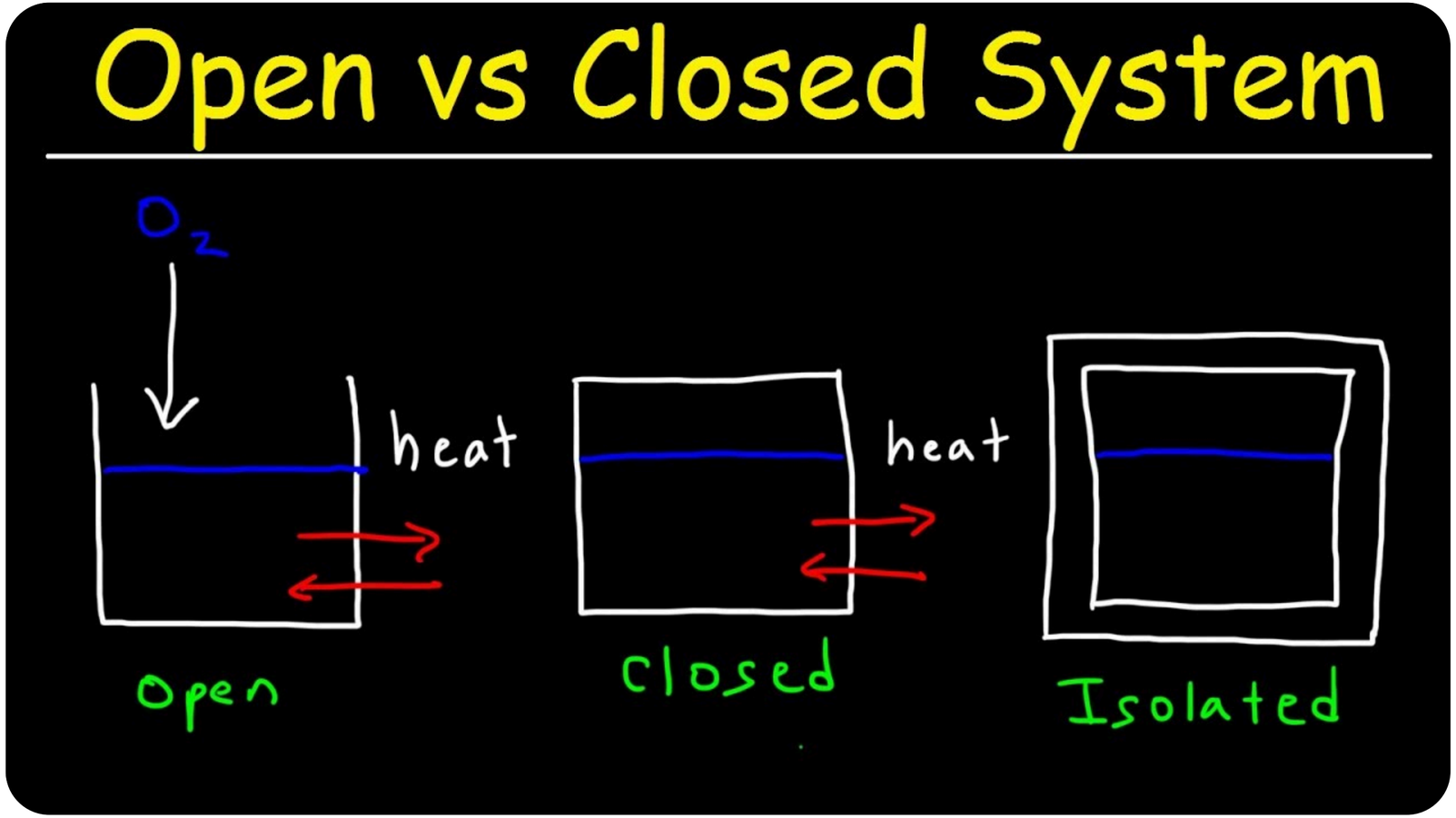
In the late 1700s, Antoine Lavoisier was the first to clearly demonstrate through experiments that mass is conserved in chemical reactions.



The Law of Conservation of Mass

Lavoisier conducted careful, quantitative experiments using a closed system so no gases or substances escaped. For example, he burned substances like phosphorus and sulfur and measured the mass before and after.

The Law of Conservation of Mass



The Law of Conservation of Mass

He disproved the Phlogiston Theory.

Flammable materials contain a fire-like element called phlogiston that is released during a combustion.



The Law of Conservation of Mass

He disproved the Phlogiston Theory.

Flammable materials contain a fire-like element called phlogiston that is released during a combustion.

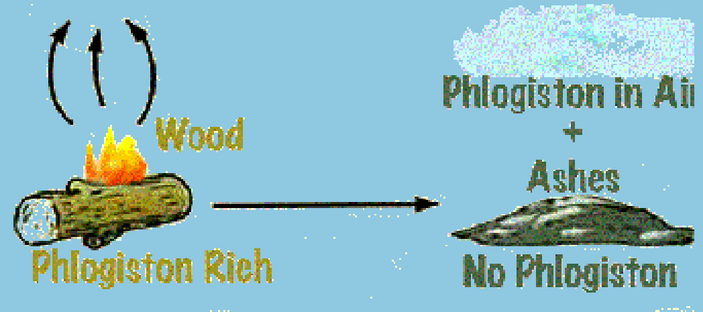


The Law of Conservation of Mass

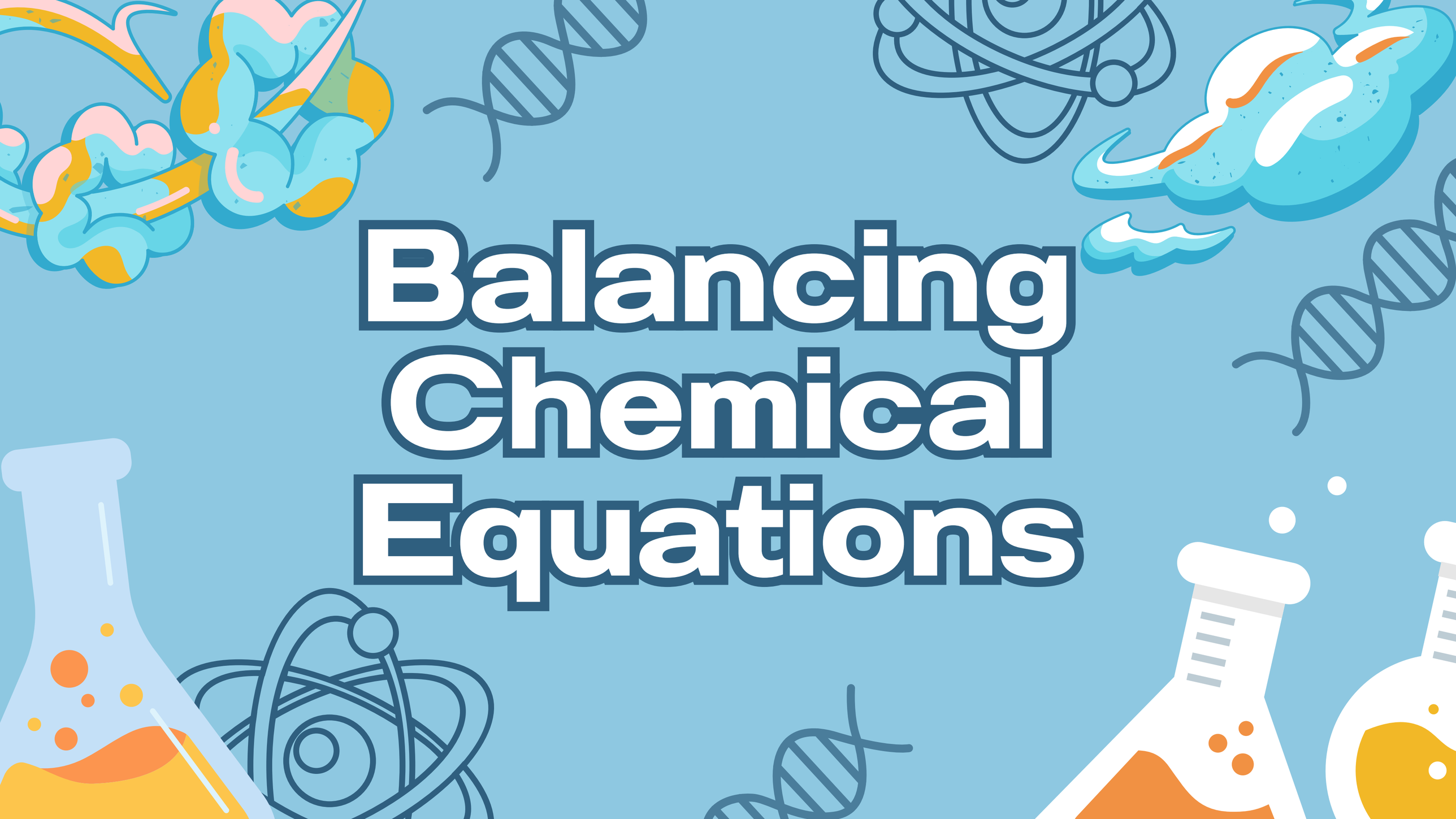


The Phlogiston Theory was proposed by Georg Ernst Stahl, a German chemist, in the late 1600s to early 1700s.

"All combustible materials contain a substance called phlogiston that is released during burning."

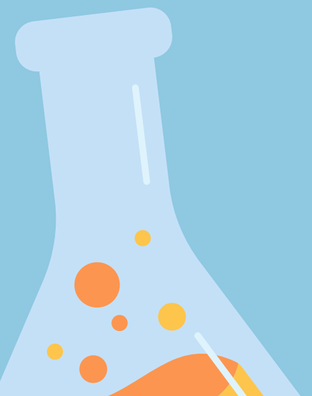


Balancing Chemical Equations



Balancing a Chemical Equation

$$H_2 + O_2 \rightarrow H_2O$$





Steps in Balancing a Chemical Equation

Step 1: Write the unbalanced equation

$$H_2 + O_2 \rightarrow H_2O$$

Steps in Balancing a Chemical Equation

Step 2: List the number of atoms of each element

$$H_2 + O_2 \rightarrow H_2O$$

H = 2	H = 2
O = 2	O = 1

Steps in Balancing a Chemical Equation

Step 3: Balance one element at a time using coefficients

Leave hydrogen and oxygen for last if they appear in multiple compounds.

Steps in Balancing a Chemical Equation

Step 4: Use coefficients to *balance*

Coefficients are the big numbers in front of formulas.

$$2H_2 + O_2 \rightarrow 2H_2O$$

Steps in Balancing a Chemical Equation

Step 5: Check all elements

Make sure each element has the same number of atoms on both sides.

The Law of Conservation of Mass

What is the connection of the Law of Conservation of Mass in balancing chemical equations?